Experiment 7: Customer Order Data Analysis Using Pandas

Aim

To analyze customer order data using Pandas DataFrame operations and find the total orders made by each customer, average order quantity for each product, and earliest and latest order dates.

Algorithm

1. Import the Pandas library.
2. Create a DataFrame containing customer IDs, order dates, product names, and order quantities.
3. Use groupby() and count() to find the total orders made by each customer.
4. Use groupby() and mean() to calculate the average order quantity for each product.
5. Convert the order date column to datetime format.
6. Use min() and max() to find the earliest and latest order dates.
7. Display the results.

Code

```
import pandas as pd

# Create order data

order_data = pd.DataFrame({

  "Customer_ID": [101, 102, 101, 103, 102, 101],

  "Order_Date": [

    "2026-10-01", "2026-10-02", "2026-10-03",

    "2026-10-04", "2026-10-05", "2026-10-06"

  ],

  "Product_Name": [

    "Laptop", "Mouse", "Mouse", "Keyboard",

    "Laptop", "Keyboard"

  ],

  "Order_Quantity": [2, 3, 4, 2, 1, 5]

})
```

```
# Convert dates to datetime
order_data["Order_Date"] = pd.to_datetime(
    order_data["Order_Date"]
)
# 1. Total orders made by each customer
total_orders = order_data.groupby("Customer_ID").size()
print("Total orders by each customer:")
print(total_orders)
# 2. Average order quantity for each product
average_quantity = order_data.groupby(
    "Product_Name"
)["Order_Quantity"].mean()
print("\nAverage order quantity for each product:")
print(average_quantity)
# 3. Earliest and latest order dates
print("\nEarliest order date:", order_data["Order_Date"].min().date())
print("Latest order date:", order_data["Order_Date"].max().date())
```

Input

Customer_ID	Order_Date	Product_Name	Order_Quantity
101	2026-10-01	Laptop	2
102	2026-10-02	Mouse	3
101	2026-10-03	Mouse	4
103	2026-10-04	Keyboard	2
102	2026-10-05	Laptop	1
101	2026-10-06	Keyboard	5

Output

```
Total orders by each customer:
Customer_ID
```

101 3

102 2

103 1

Average order quantity for each product:

Product_Name

Keyboard 3.5

Laptop 1.5

Mouse 3.5

Earliest order date: 2026-10-01

Latest order date: 2026-10-06

```
====== RESTART: C:/Users/sivak/Desktop/FODS LAB experiments codes/exp 7.py =====
Total orders by each customer:
Customer_ID
101    3
102    2
103    1
dtype: int64

Average order quantity for each product:
Product_Name
Keyboard    3.5
Laptop      1.5
Mouse       3.5
Name: Order_Quantity, dtype: float64

Earliest order date: 2026-10-01
Latest order date: 2026-10-06
>
```

Result

Thus, the total orders per customer, average order quantity per product, and earliest and latest order dates were successfully determined using Pandas.